Brain MRI Tumor Segmentation and Quantitative Analysis Using Classical Image Processing

Week 6 - Further Ideation and Scaling based on Python

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I. Work done this Week:-

This week we looked into our plan post the mid semester review, with further discussion with Prof. Mehul Raval, we learnt that instead of causing high computation by using the entire scan to detect this tumor within the scan, but when we tried to change the code back to work on individual slices, so instead we have chosen to change our base scan, and went through multiple iterations on the same to better understand the process, as each new base would give different readings on each individual scan.

In Other Plans, We have further looked into everything we have to preprocess in Python before moving on the C++, this basically includes improvement on the subtraction process, basically doing it after we downscale the image from float64 to uint16 instead of directly trying to apply it on the float64 scan, we are also further trying to balance out the scaling for the scans. Another Major concern that is showing up for us is the fact that the scans we are using are all of different dimensions due to the constraint of time, basically the period of scan for each individual patient

We analyzed our MRI subtraction pipeline and identified several issues affecting the alignment and quality of the output. Our existing preprocessing code resampled the scans to a common voxel grid without performing actual anatomical registration, resulting in differences in brain position and orientation. We also found inconsistencies in voxel-axis ordering and dimension handling during binary file export, which could cause the voxel data to be interpreted incorrectly. Additionally, our brain extraction method relied on thresholding and morphological operations, which could leave unwanted non-brain regions in the scans. The main issue with the code is that we have taken the first scan as a baseline check so the scans which are processed after that are subtracted with the reference of the first scan, we have changed it and added a target scan selection by user which can be used as a reference.

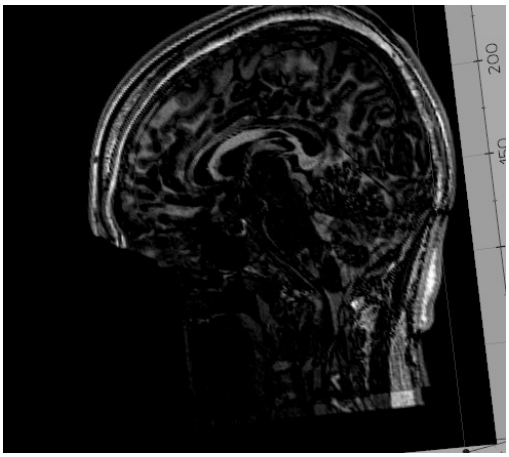


Fig. 1. misaligned result

We also identified that each MRI scan was independently normalized to the full 16-bit intensity range, making intensity values less comparable across scans. Furthermore, using the first scan's affine geometry with a fixed target shape could lead to incorrect spatial placement or cropping. These issues collectively affected the reliability of the voxel-wise subtraction and contributed to the misaligned output image.

Our current progress in python involves scaling and aligning the different dimension images. As the images are not all of the same size, we have done one approach of scaling and aligning the images, computed in python. This has resulted in

partial success as the images are about scale as we expect. Our subtraction test using CUDA C++ also shows that most of the brain part is being subtracted from the baseline images accurately shown by the image.

II. Work to be done Next Week:-

As Priorly Mentioned, We have thought up a new plan for the next few weeks, We are going to complete any portion remaining to be resolved in Python and then change our programming language to c++, there we will again have to deal with concepts of loading the image, and baseline formation on the preupdated scan surrounding it, for which we are planning to use CUDA. We are further going to test the data against a baseline and then actually work further towards detecting the tumor within the scan, and then we are going to export this result from teh current standpoint.

We again do realise that working with the image would be much easier to deal with in Python, so we will return to Python for our final proceedings, so we would load back our updated image into Python and rerun all the test specific to Python to make sure that our results have worked accurately or not. We are then gonna work on highlighting the specific tumor regions present within multiple patients scan/

At around the end of the project, we plan to create an interactive GUI, for Doctors and Patients to use, so that they have a point of reference when looking at MRI scans to get a deeper and more meaningful explanation behind their condition, this will be more visible compared to how it's being done right now. But currently looking at our current work, we have to deal with all the issues that have occurred this week, try to further correct them and hope to have a smooth transition when changing our language for better processing.

Repo Link:-
https://github.com/Rushin-Mod/DIP_2026_P20_G9↗